

|       | f1  | f2  | f3  |
|-------|-----|-----|-----|
| row 1 | 30  | 60  | NaN |
| row 2 | NaN | 55  | 20  |
| row 3 | 40  | 52  | 25  |
| row 4 | 25  | NaN | 22  |
| row 5 | 50  | 70  | 40  |

fill f1 of row 2

**univariate: column mean**

only f1 is used

$$(30 + 40 + 25 + 50)/4 = 36.25$$

**multivariate: KNN imputer**

f2 and f3 find the 2 most similar rows (4 and 3):  $(25 + 40)/2 = 32.5$